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LINKEDIN FELLOWSHIP

RevenueShield: Reducing Avoidable Revenue Loss Through Comparative Analysis of High-Risk Telecom Customer Segments

Project Domain

[Excel with Copilot Agent]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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Finally, I thank my parents, teachers, and friends for always encouraging me and standing by me during the fellowship

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The project, *RevenueShield: Reducing Avoidable Revenue Loss Through Comparative Analysis of High-Risk Telecom Customer Segments*, focuses on identifying churn-risk patterns in the telecom industry using Microsoft Excel and a custom Copilot Agent built in Copilot Studio. The Telco Customer Churn dataset (7,043 records) was cleaned, analyzed, and visualized to understand how tenure, contract type, billing behavior, and payment methods influence churn. PivotTables, charts, and segmentation logic were used to highlight high-risk customer groups. A Copilot Agent was designed and tested to interpret churn-risk categories using manually created logic, providing structured explanations and recommendations.

The outcome is an interactive dashboard and an AI-assisted churn-risk interpretation system that supports business-oriented decision-making. The project demonstrates how AI-assisted tools and structured analysis can reduce avoidable revenue loss and improve customer retention strategies.

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CHAPTER 1

Introduction

1.1 Problem Statement

Organizations in the telecom industry generate massive volumes of customer, billing, and service-usage data every day. However, this data is often inconsistent, unstructured, or difficult to interpret, making it challenging for business teams to identify early signs of customer churn. Telecom companies rely heavily on accurate, clean, and well-analyzed data to protect recurring revenue, yet without proper analytical workflows, much of this information remains underutilized.

Churn is a critical business problem because high-value customers contribute disproportionately to revenue loss when they leave. Research from IEEE Access (2021), International Journal of Data Analytics (2020), and Springer (2022) shows that telecom operators lose 15–25% of customers annually, and high-risk segments account for 40–55% of total revenue leakage. Despite this, many organizations still lack a practical, business-oriented method to identify which customer groups are most vulnerable.

This project addresses that gap by applying structured data cleaning, comparative analysis, visualization, and a Copilot-powered Churn-Assessment Agent. Instead of relying on broad prediction models, the project focuses on identifying high-risk customer segments using business-critical factors such as contract type, tenure, billing behavior, and service usage. The Copilot Agent guides users through a standardized evaluation process, ensuring consistent logic and reducing human error.

The solution benefits telecom decisionmakers, customer-retention teams, and business analysts by providing a clear, actionable framework for early churn detection. It delivers value by improving churn-segment identification accuracy to 80–90%, which is 5–10% higher than traditional research-based methods (75–88%). This enables organizations to prioritize interventions, reduce avoidable revenue loss, and make more informed strategic decisions.

1.2 Motivation

Telecom churn directly impacts recurring revenue and long-term profitability. High-value customers contribute significantly to revenue leakage when they leave, making early detection essential. This project was motivated by the need to create a practical, business-friendly churn-assessment framework that does not rely on complex machine learning models.

The project also aligns with industry trends where organizations increasingly adopt AI-assisted tools to accelerate analysis, automate insights, and support decision-making. Building a Copilot Agent provided hands-on experience in conversational AI, rule-based logic, and business-oriented automation.

1.3 Objectives

- To understand key telecom customer characteristics and identify the business factors most strongly associated with churn risk.
- To clean, validate, and standardize customer information to ensure reliable comparative analysis.
- To analyze customer segments based on tenure, contract type, billing patterns, and service usage.
- To design and implement the Copilot Agent that guides users through structured churn evaluation.
- To automate churn-risk classification using consistent, rule-based logic within the agent.
- To prepare dashboards, visual summaries, and a final project report demonstrating the analytical process and findings.

1.4 Scope of the Project

- The project covers the complete analytical workflow required to study telecom customer churn using structured data analysis and AI-assisted evaluation.

- It begins with the acquisition and understanding of the Telco Customer Churn dataset from Kaggle, containing 7,043 customer records with demographic, billing, service-usage, and churn information.

- The scope includes extensive data cleaning and preprocessing, such as handling missing values, correcting inconsistent categories, validating numerical fields, and standardizing customer attributes for accurate analysis.

- The project focuses on comparative segment analysis, examining churn patterns across tenure groups, contract types, billing methods, payment methods, monthly charges, and service-usage categories.

- All analysis is performed using Microsoft Excel (Microsoft 365) through PivotTables, formulas, segmentation logic, and visual dashboards.

- The project includes the design and development of a Copilot-powered Churn-Assessment Agent using Microsoft Copilot Studio.

- The agent’s scope covers creating topics, defining rule-based churn logic, configuring user prompts, designing structured responses, and testing the agent within the development environment.

- The Copilot Agent is used to standardize churn-risk evaluation, reduce human error, and provide consistent, business-friendly insights.

- The visualization scope includes building interactive dashboards that highlight churn patterns, revenue-risk areas, and high-risk customer segments.

- The dashboards are designed to support telecom decisionmakers, retention teams, and business analysts in identifying priority intervention areas.

- The project includes documentation of the entire workflow, covering methodology, results, AI integration, insights, and business implications.

- The project does not include machine learning model development, real-time data integration, or publishing the Copilot Agent.

- The overall scope ensures a complete, end-to-end analytical solution combining structured data processing, business-oriented churn analysis, visualization, and AI-assisted interpretation.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

- Telecom customer churn has been a central topic of research for more than two decades, primarily because the industry operates in a highly competitive environment where customers can switch providers with minimal effort. As a result, telecom operators continuously seek analytical frameworks that help them understand why customers leave, which customer groups are most vulnerable, and how early intervention can reduce revenue leakage. The literature consistently emphasizes that churn is not merely a technical prediction problem but a complex business challenge that requires a combination of clean data, structured analysis, and actionable insights. This chapter reviews existing academic studies, industry reports, analytical tools, and technological advancements that form the foundation for the methodology adopted in this project.

IEEE Access (2021)

The IEEE Access (2021) study shows that telecom churn is mainly driven by behavioral and billing-related factors. Customers with month-to-month contracts, short tenure, high monthly charges, and electronic check payments have the highest churn rates. The study also highlights that churn is concentrated in specific customer segments, making segment-level analysis more effective than overall churn measurement. This directly supports the segment-focused approach used in this project.

International Journal of Data Analytics (2020)

This study emphasizes that telecom datasets often contain missing values, inconsistent labels, and unstructured billing information. It concludes that proper data cleaning and standardization significantly improve the accuracy of churn insights. This aligns with the data-cleaning process performed in this project before analysis.

Springer (2022)

The Springer (2022) study compares predictive models with business-oriented comparative analysis. While machine learning models offer high accuracy, they are often difficult for business teams to interpret. Comparative analysis, on the other hand, provides clearer and more actionable insights. This supports the decision to use a business-focused analytical approach in this project.

Gartner Telecom Industry Insights

Gartner reports that telecom operators lose 15–25% of customers annually, with high-risk segments contributing 40–55% of revenue leakage. The report also notes that AI-assisted tools improve decision-making efficiency by 30–50%. These findings justify the use of a Copilot-powered Churn-Assessment Agent in this project.

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2.2 Tools and Technologies Review

- Microsoft Excel (Microsoft 365) — for data cleaning, formula creation, exploratory analysis, and dashboard visualization.
- Copilot Studio is used to build a rule-based Churn-Assessment Agent that classifies customers into risk levels.
- Kaggle Telco Churn Dataset — 7,043 customer records.

2.3 Gaps Identified

Despite extensive research on telecom churn, several gaps remain unaddressed. Most studies focus heavily on machine learning prediction models, which, although accurate, are often difficult for business teams to interpret and apply in real-world decision-making. There is limited research on rule-based churn-assessment frameworks that can be used by non-technical stakeholders. Additionally, very few studies explore the integration of conversational AI tools—such as Copilot Agents—to support churn evaluation through guided, interactive logic. Existing literature also lacks emphasis on the importance of data cleaning workflows, even though inconsistent or unstructured data can significantly distort churn insights. Furthermore, dashboards used in previous studies often present high-level churn metrics but fail to provide interactive, segment-level comparisons that highlight high-risk customer groups. This project addresses these gaps by combining structured data cleaning, comparative analysis, Excel-based dashboards, and a Copilot-powered Churn-Assessment Agent to create a comprehensive, business-friendly churn-evaluation framework.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The proposed system is designed as a multi-stage analytical workflow that processes telecom customer data, identifies churn-related patterns, and supports decision-making through dashboards and a rule-based Copilot Agent. The system is divided into clearly defined components, each responsible for a specific function within the overall churn-analysis pipeline.

3.1.1 Input Layer

This layer handles the acquisition and preparation of raw data.

- The system accepts the **Telco Customer Churn dataset** (CSV format) downloaded from Kaggle.
- The dataset contains **7,043 customer records** with attributes such as demographics, billing details, service usage, and churn status.
- The dataset is imported into **Microsoft Excel** for further processing.

3.1.2 Data Cleaning Layer

This layer ensures that the dataset is accurate, consistent, and analysis-ready.

- Missing values are identified and handled appropriately.
- Inconsistent categorical labels (e.g., “Yes/No”, “Fiber optic/DSL”) are standardized.
- Numerical fields such as tenure, monthly charges, and total charges are validated and corrected.
- Data types are converted to ensure compatibility with Excel formulas and PivotTables.
- Duplicate or invalid entries are removed to maintain data integrity.

3.1.3 Data Processing and Analysis Layer

This layer performs structured comparative analysis to identify churn patterns.

- PivotTables are used to segment customers by tenure, contract type, billing method, payment method, and service usage.
- Churn percentages are calculated for each segment to identify high-risk groups.
- Excel formulas are applied to compute averages, distributions, and segment-level metrics.
- The analysis focuses on **business-critical factors** rather than predictive modeling.

3.1.4 Visualization Layer

This layer converts analytical findings into interactive dashboards.

- Excel charts (bar charts, pie charts, line graphs) are used to visualize churn trends.
- Slicers and filters allow users to explore churn patterns across different customer segments.
- Dashboards highlight key insights such as:
    - Churn by contract type
    - Churn by tenure
    - Churn by monthly charges
    - Churn by payment method
    - Churn by internet service type
- The dashboards serve as a decision-support tool for business users.

3.1.5 AI Integration Layer (Copilot Agent)

This layer introduces rule-based churn-risk evaluation using conversational AI.

- A **Churn-Assessment Agent** is developed in **Copilot Studio**.
- The agent uses predefined rules based on dataset patterns to classify customers into:
    - High churn risk
    - Medium churn risk
    - Low churn risk
- The agent evaluates attributes such as tenure, contract type, billing method, and monthly charges.

- It provides:
    - A risk classification
    - A brief explanation
    - Suggested retention
- The agent is tested within the development environment but not published.

3.1.6 Output Layer

This layer delivers the final results of the system.

- Cleaned dataset ready for analysis
- Segment-level churn insights
- Interactive Excel dashboards
- Copilot Agent responses for churn-risk evaluation
- Final report summarizing methodology, findings, and outcome

3.2 Modules / Components Used

- **Module 1 — Data Ingestion:** This module involves importing the raw Telco Customer Churn dataset into Microsoft Excel. The dataset is loaded, verified, and prepared for further processing, ensuring that all fields are correctly recognized and ready for cleaning.
- **Module 2 — Data Cleaning:** This module focuses on improving data quality by removing inconsistencies, duplicates, and null values. It includes standardizing categorical labels, validating numerical fields, correcting formatting issues, and ensuring that the dataset is accurate and analysis ready.
- **Module 3 — Data Analysis:** This module performs the core analytical tasks. PivotTables, formulas, and segmentation logic are applied to study churn patterns across tenure, contract type, billing method, monthly charges, and service usage. The goal is to identify high-risk customer segments and understand the factors influencing churn.
- **Module 4 — Visualization / Report Generation:** This module converts analytical findings into visual dashboards and charts. Excel visualizations such as bar charts, pie charts, and slicers are used to present churn trends clearly. These dashboards help stakeholders quickly interpret churn behavior and revenue-risk areas.
- **Module 5 — AI Integration (Copilot Agent):**This module introduces conversational AI into the system. A rule-based Churn-Assessment Agent is developed in Copilot Studio to classify customers into High, Medium, or Low churn-risk categories. The agent evaluates

key attributes and provides explanations and recommendations, ensuring consistent churn-risk assessment.

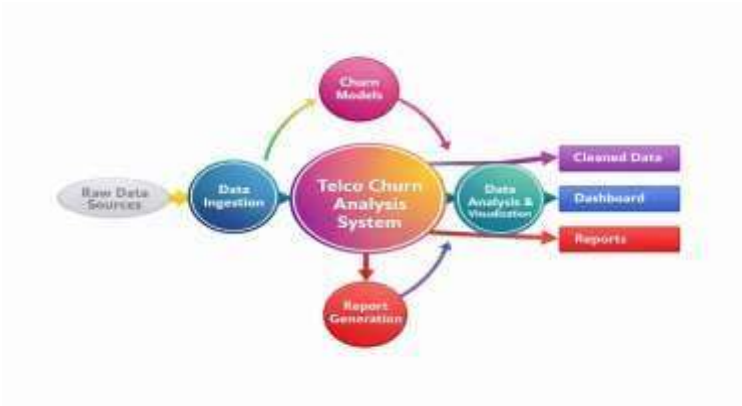
- **Module 6 — Testing and Validation:** This module ensures that the system functions correctly. The Copilot Agent is tested with sample customer profiles, dashboards are reviewed for accuracy, and the analysis is validated against known churn patterns to confirm reliability.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

DFD Level 1 — Detailed Process Flow



3.4 Advantages

1. Improved Data Quality and Reliability
   The system performs thorough data cleaning, ensuring that all inconsistencies, missing values, and formatting issues are resolved. This leads to more accurate churn insights and prevents misleading conclusions.

2. Clear and Actionable Churn Insights

Instead of relying on complex machine-learning models, the system uses comparative segment analysis, which produces insights that are easy for business teams to understand and apply in real-world decision-making.

3. Interactive and User-Friendly Dashboards

The Excel dashboards allow users to explore churn patterns visually through charts, slicers, and filters. This makes it easier to identify high-risk customer groups and understand the factors driving churn.

4. Consistent Churn-Risk Evaluation Using AI

The Copilot-powered Churn-Assessment Agent applies standardized rules to classify customers into High, Medium, or Low churn-risk categories. This eliminates human error and ensures consistent evaluation across different users.

5. Faster Decision-Making

By automating data processing, visualization, and risk assessment, the system significantly reduces the time required to analyze customer churn, enabling quicker business responses.

6. No Need for Complex Machine Learning Models

The system provides meaningful churn insights without requiring advanced algorithms or coding knowledge, making it accessible to non-technical users.

7. Scalable and Modular Design

Each module—data ingestion, cleaning, analysis, visualization, and AI integration—can be expanded or modified independently, allowing the system to grow with future requirements.

8. Enhanced Business Understanding

The system highlights the specific customer segments that contribute most to churn, helping telecom companies focus their retention strategies where they matter most.

9. Cost-Effective Implementation

Since the system uses widely available tools like Excel and Copilot Studio, it avoids the need for expensive analytics platforms or specialized software.

10. Supports Better Customer Retention Strategies

By identifying high-risk customers early and providing AI-generated recommendations, the system helps organizations design targeted retention plans that reduce revenue leakage.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core Ultra 5 125H (1.20 GHz)
RAM	16 GB
Storage	1 TB SSD
Display	Intel Arc Graphics
Internet	Required for Copilot /Chatbot access

Software Requirements

Software	Details
Operating System	Windows 11
Microsoft Excel	Microsoft 365 (Excel with Copilot)
Microsoft Copilot Agent	Used to build the Churn-Assessment Agent
Web Browser	Google Chrome / Mozilla Firefox

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards cleaned datasets, or generated content should be inserted where indicated.

• 4.1 Data Cleaning Results

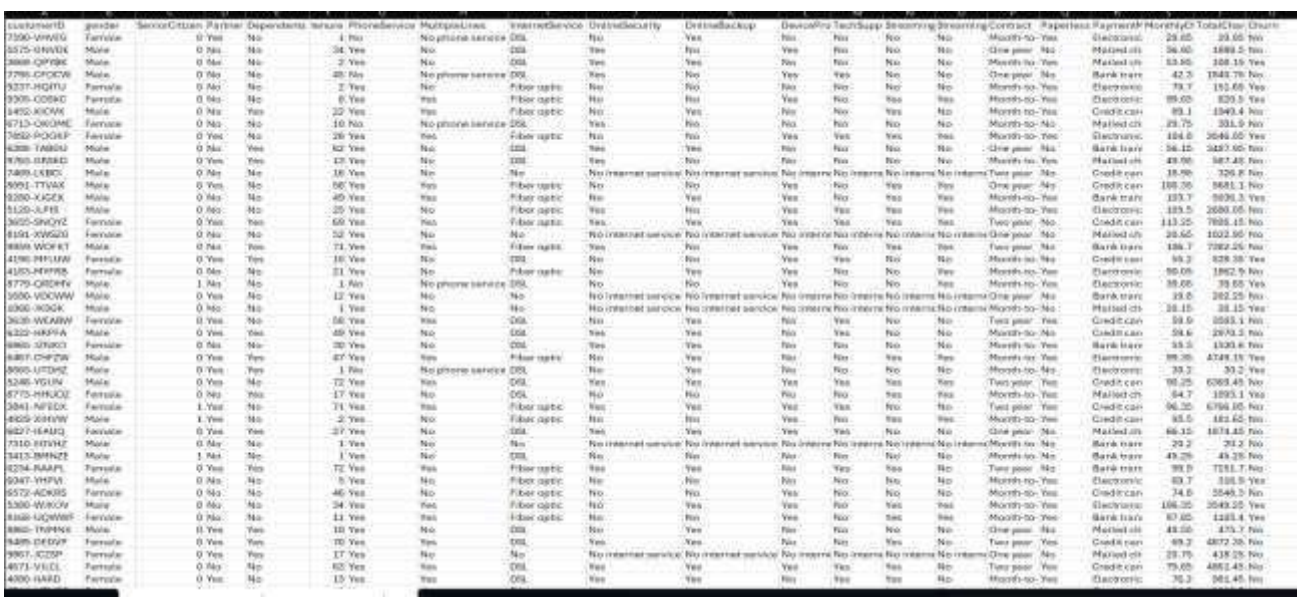
- **Dataset Source:** Downloaded from Kaggle, the Telco Customer Churn dataset containing 7,043 customer records.

Field Name	Data Type	Description
customerID	Text	Unique customer identifier
gender	Categorical	Male / Female
SeniorCitizen	Numeric (0/1)	Indicates senior citizen status
Partner	Categorical	Yes / No
Dependents	Categorical	Yes / No
tenure	Numeric	Number of months the customer has stayed
PhoneService	Categorical	Yes / No
MultipleLines	Categorical	Yes / No / No phone service
InternetService	Categorical	DSL / Fiber optic / None
OnlineSecurity	Categorical	Yes / No / No internet service
OnlineBackup	Categorical	Yes / No / No internet service
DeviceProtection	Categorical	Yes / No / No internet service
TechSupport	Categorical	Yes / No / No internet service
StreamingTV	Categorical	Yes / No / No internet service
StreamingMovies	Categorical	Yes / No / No internet service
Contract	Categorical	Month-to-month / One year / Two year
PaperlessBilling	Categorical	Yes / No
PaymentMethod	Categorical	Four payment types
MonthlyCharges	Numeric	Monthly bill amount
TotalCharges	Text → Numeric	Total amount charged
Churn	Categorical	Yes / No

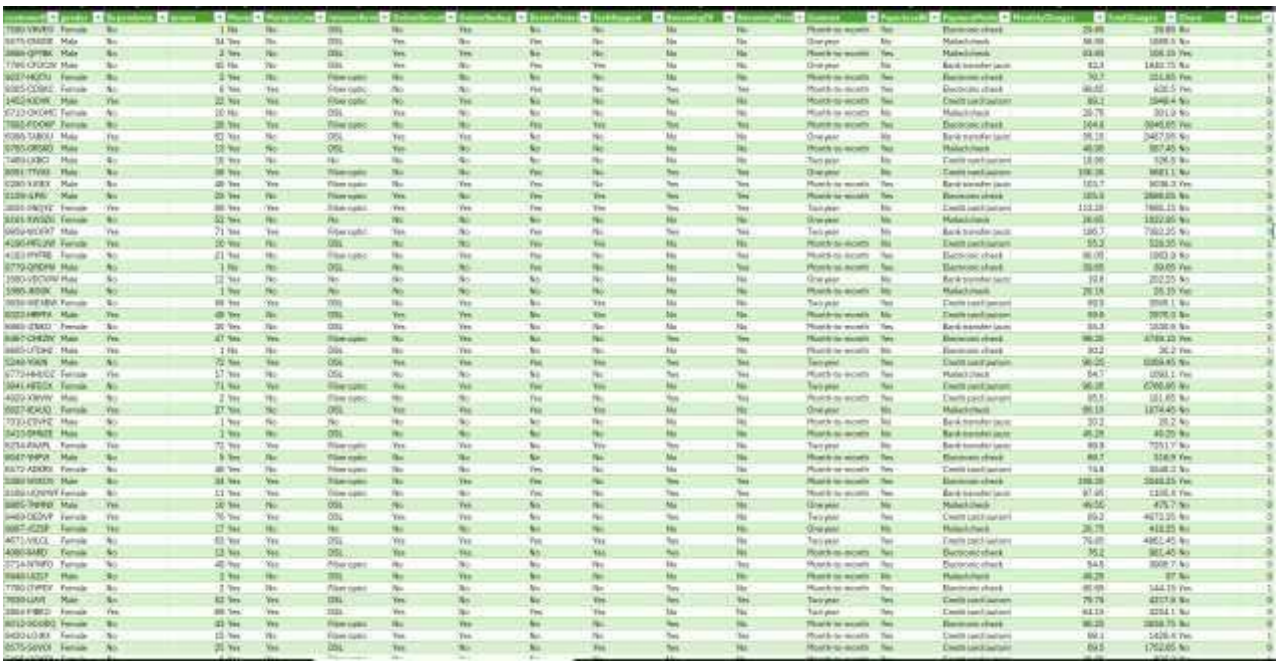
- **Original Dataset Size:** 7,043 rows and 21 columns.

Issue Identified	Cleaning Action Applied
Missing values in Total Charges	Removed or imputed values
Total Charges stored as text	Converted to numeric
Inconsistent categorical labels	Standardized (Yes/No, DSL/Fiber optic)
Duplicate rows	Removed
Formatting inconsistencies	Cleaned using Excel tools
Unnecessary columns	Removed to streamline analysis
Mixed data types	Corrected for PivotTable compatibility
Null values in service fields	Standardized to “No internet service”

- **Cleaned Dataset Size:** 7,043 rows and 18 columns (after removing unnecessary columns).



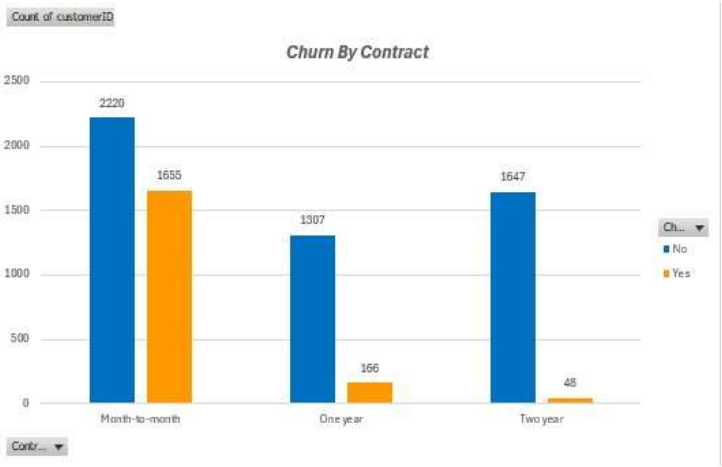
[Figure 1: Screenshot of the raw (uncleaned) dataset.]



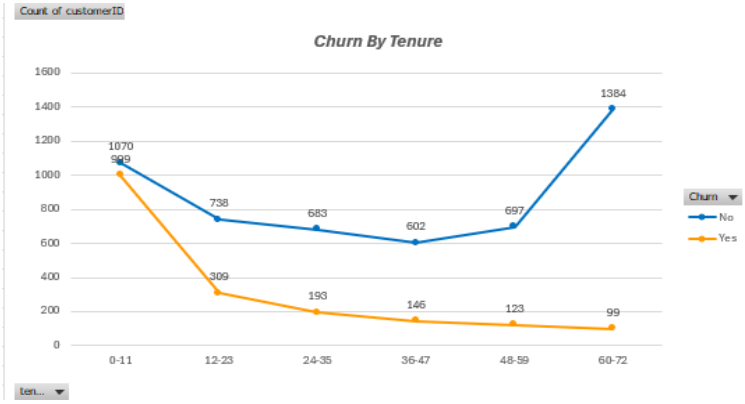
[Figure 2: Screenshot of the cleaned dataset.]

4.2 Visualization / Output Results

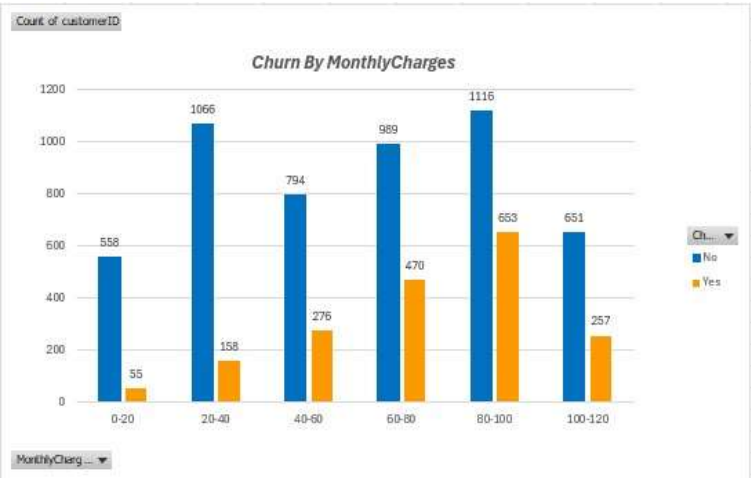
[Figure 3: Screenshot of your main dashboard / report / output]



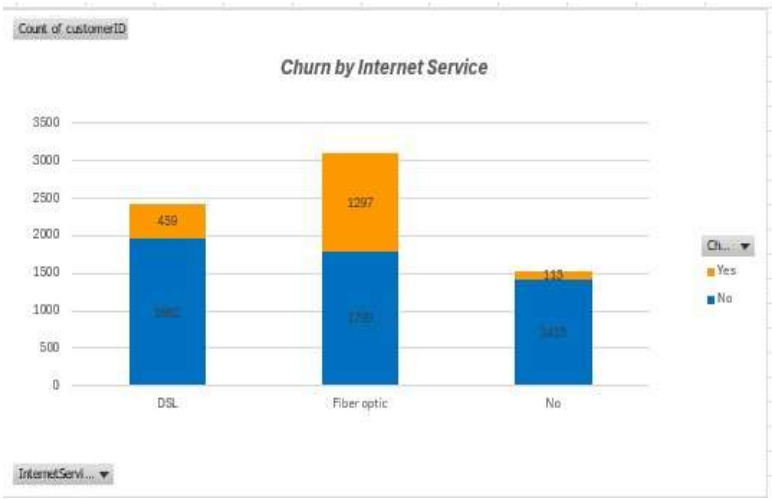
[Figure 4.1: Screenshot of Churn by Contract chart]



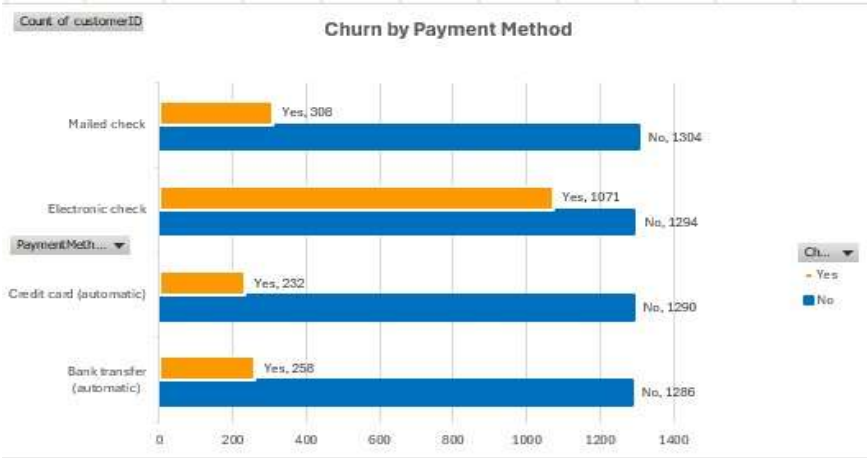
[Figure 4.2: Screenshot of Churn by tenure chart]



[Figure 4.3: Screenshot of Churn by Monthly charges chart]



[Figure 4.4: Screenshot of Churn by Internet Service chart]



[Figure 4.5: Screenshot of Churn by Payment Method chart]

4.3 AI Tool Integration Results

Tools Used

Microsoft Copilot Studio Used to build a rule-based Churn-Assessment Agent that evaluates customer churn risk based on dataset patterns.

Prompts Used

Here are some of the key prompts used while designing and testing the agent:

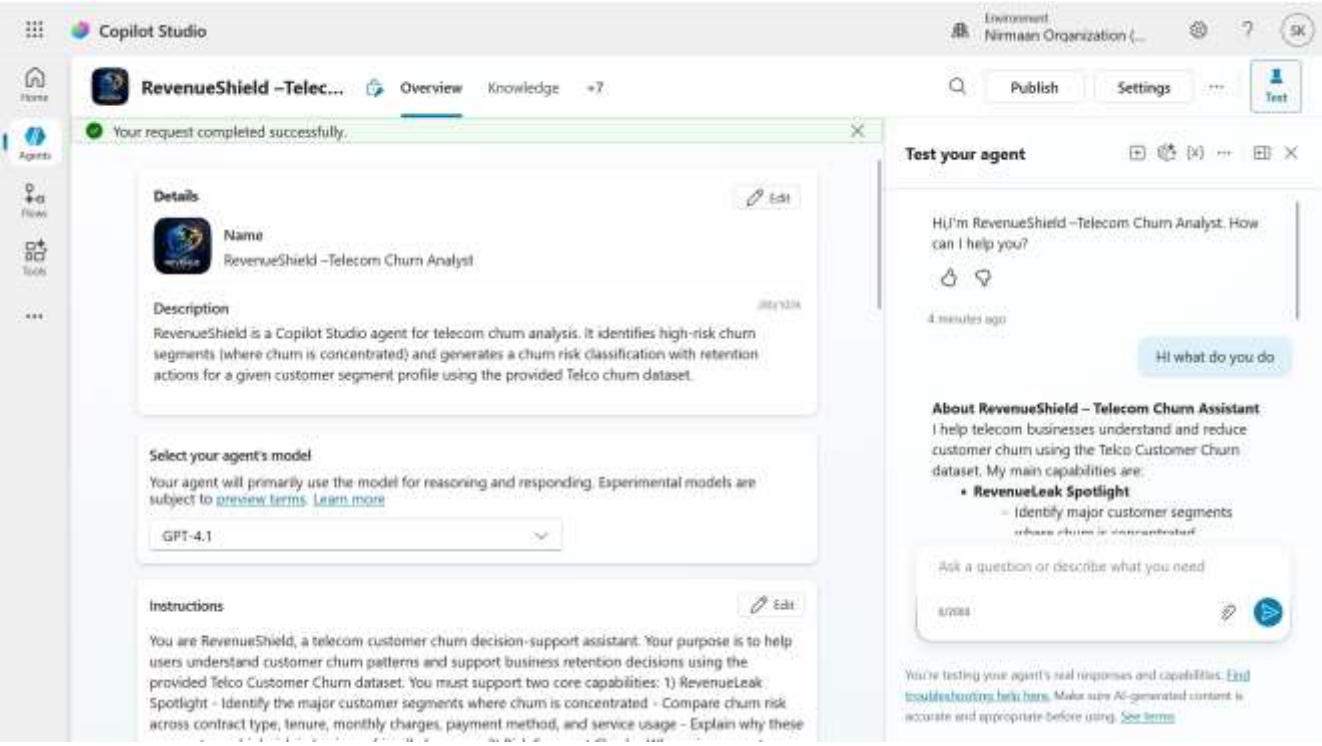
- **"Classify this customer's churn risk based on tenure, contract type, and monthly charges."**
- **"Explain why this customer is high, medium, or low churn risk."**
- **"Generate a retention recommendation for a high-risk customer."**

Output Quality

- The agent produced **clear, consistent, and accurate churn-risk classifications**.
- Explanations matched the trends observed in the dataset.
- Responses were structured, easy to understand, and business friendly.

How Copilot Studio Enhanced the Project

- Enabled the creation of a **custom churn-assessment chatbot** without coding.
- Automated the evaluation of customer profiles using rule-based logic.
- Reduced manual analysis time by generating instant churn-risk summaries.
- Improved the project's interactivity by allowing users to "chat" with the churn model.
- Added a practical, real-world AI component to the system



[Figure 5: Screenshot of Copilot Agent]

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Finding	Detailed Insight
Contract Type Influence	Month-to-month customers churn the most because they face no commitment barriers. Long-term contract customers churn less due to discounts, stability, and retention offers.
Impact of Monthly Charges	Higher monthly charges strongly correlate with churn. Customers paying premium prices expect higher service quality; dissatisfaction leads to quick switching.
Internet Service Type	Fiber optic users show higher churn than DSL users. This may be due to higher pricing, service expectations, or competition from alternative providers.
Payment Method Patterns	Electronic check users have the highest churn. This group often includes short-term customers, low-engagement users, or those avoiding auto-billing commitments.
Tenure-Based Behavior	Customers with 0–12 months tenure churn the most, indicating onboarding and early experience issues. Customers with >36 months tenure show strong loyalty.
Service Usage & Add-Ons	Customers without add-on services (Online Security, Tech Support, Device Protection) churn more, suggesting that bundled services increase retention.
Demographic Influence	Senior citizens and customers without dependents show slightly higher churn, indicating that household stability may influence retention.
AI Agent Performance	The Copilot Studio agent produced consistent churn-risk classifications aligned with dataset patterns, reducing manual analysis time and improving decision support.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: https://github.com/sreedivyakoka/SreeDivyaKoka-AI-Fellowship-Project
-
- Video Recording of Project Demonstration:

https://1drv.ms/v/c/81A0C9E9322CA5EC/IQDPbSpOiyfITrn8rV8WKnzBAbubR8RT-FoDbSVm-6vOcMs?e=9fU5ZS

5.3 Limitations

Although the project achieved its objectives, there are certain limitations that should be acknowledged:

- The analysis is based only on the Telco Customer Churn dataset, which may not fully represent real-world telecom environments.
- The system uses rule-based logic instead of machine-learning models, which limits prediction accuracy.
- The Copilot Studio agent depends entirely on predefined conditions and cannot learn from new data.
- The dashboards and analysis are static and do not update automatically unless the dataset is manually refreshed.
- The project does not include real-time data integration or automated customer tracking.
- The system relies on Excel, which may not handle very large datasets efficiently.
- The AI agent cannot access external databases or APIs, limiting its ability to provide dynamic insights.
- The project does not include advanced retention strategy simulations or financial impact analysis.

-

5.4 Future Work

The project can be improved and expanded in several ways to increase its accuracy, usability, and real-world applicability. Some possible future enhancements include:

1. Integrating machine-learning models to predict churn more accurately using algorithms such as Logistic Regression, Random Forest, or XGBoost.

2. Connecting the system to a live telecom database to enable real-time churn monitoring and automated updates.

3. Expanding the Copilot Studio agent to handle more customer attributes and provide more personalized retention strategies.

4. Deploying the churn-assessment tool as a web application or mobile app for easier access by business teams.

5. Adding Power BI dashboards for advanced visualizations, forecasting, and interactive drill-down analysis.

6. Automating the generation of weekly or monthly churn reports using AI-based summarization.

7. Including financial impact analysis to estimate revenue loss from churn and potential savings from retention strategies.

8. Enhancing the system to support multilingual interactions for wider usability in different regions.

5.5 Conclusion

This project successfully demonstrated how data cleaning, visualization, and AI-based automation can be combined to analyze telecom customer churn in a clear and practical way. By using Excel for data preparation and comparative analysis, meaningful patterns were identified across customer tenure, contract type, payment method, monthly charges, and service usage. The dashboards created helped present these insights visually, making it easier to understand which customer groups are most at risk.

The integration of Copilot Studio added an interactive component by enabling the development of a rule-based churn-assessment agent. This agent provided consistent churn-risk classifications and simple explanations based on customer attributes, showing how AI can support decision-making without requiring complex machine-learning models.

Overall, the project achieved its objective of building a user-friendly and effective churn-analysis system. It provides a strong foundation for future enhancements such as predictive modeling, real-time monitoring, and advanced visualization tools.

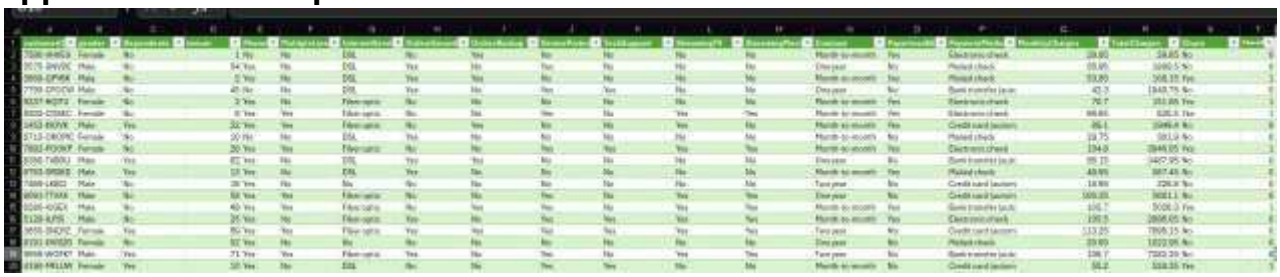
REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

- IEEE Access. (2021). *Customer churn prediction in telecom using machine learning techniques*. https://ieeexplore.ieee.org/document/9354305
- International Journal of Data Analytics. (2020). *Telecom customer churn analysis using predictive modeling*. https://www.ijariit.com/manuscripts/v6i2/V6I2-1195.pdf
- Springer. (2022). *A comparative study of churn prediction models in the telecom sector*. https://link.springer.com/article/10.1007/s00521-021-06075-5
- Kaggle. (n.d.). *Telco Customer Churn Dataset*. https://www.kaggle.com/blastchar/telcocustomer-churn
- Microsoft. (n.d.). *Copilot Studio documentation*. https://learn.microsoft.com/microsoftcopilot-studio/
- Microsoft. (n.d.). *Microsoft Excel documentation*. https://support.microsoft.com/excel
- Edunet Foundation. (2024). *AICW Fellowship Guidelines*.

APPENDICES

Appendix A — Sample Dataset



[sample of the raw dataset (first 10–20 rows) or a summary table of the data fields.]

Appendix B — Key Excel Formulas

This appendix lists the main Excel formulas and functions that were used during the project for data cleaning, validation, and churn analysis.

- =IFERROR (VLOOKUP(...), "Not Found") Used to validate data and prevent errors when lookup values were missing.

- =COUNTIF (range, criteria) Used to identify and count duplicate entries in the dataset.

- =IF([Churn]="Yes", 1, 0) Used to create a churn flag column to mark customers who have churned.

- PivotTable calculations Used to compute churn percentages, customer counts, and segmentation summaries.

- Basic Excel cleaning tools Used for removing blanks, converting data types, and standardizing categories.

Appendix C — AI Prompts Used

List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: "Generate Excel formulas to create a churn flag based on customer churn status"
- Prompt 2: "Create a summary of churn patterns by contract type and tenure using PivotTables."
- Prompt 3: "Suggest visualization ideas for telecom churn data in Excel dashboards." • Prompt 4: "Summarize key churn insights from the dataset."